Development of the University of Canterbury Four-Alternative Forced Choice (UC4AFC) Monosyllabic Word List

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Introduction

- Auditory processing disorder (APD) may present as listening difficulties despite normal peripheral hearing
- The UCAST-FW (O'Beirne et al., 2012; Rickard et al., 2013) is an adaptive low-pass-filtered speech test used in APD assessment
- Original UCAST-FW stimuli (NU-CHIPS) are less suitable for NZ/Australian children
- To address this, the UC4AFC monosyllabic word list was developed using NZ/Australian English phonemic contrasts
- Initial 98-word list (created by Murray and O'Beirne; Murray, 2012) included:
  - target word
  - one semantic or unrelated foil
  - two phonemic foils
- Here we summarise three refinement and validation studies supporting paediatric clinical use

Study 1 – Recording words, selecting images (Anderson, 2024)

Objectives:

i) Select representative images for each of the 98 target words

ii) Create audiovisual recordings for future AV-integration studies

iii) Evaluate low-pass-filter intelligibility characteristics of the recordings

Method:

- 392 candidate images collected (4 per target word)
- 43 adults selected "best image for children <12 years"
- Audiovisual recordings created for all target words
- 35 adults completed auditory-alone identification under 8–9 low-pass filter conditions (200–1500 Hz)
- Responses collected in:
  - open-set format
  - closed-set four-alternative forced choice (4AFC) format
- Data used to evaluate low-pass-filter psychometric characteristics for word list normalisation *(data not shown)*

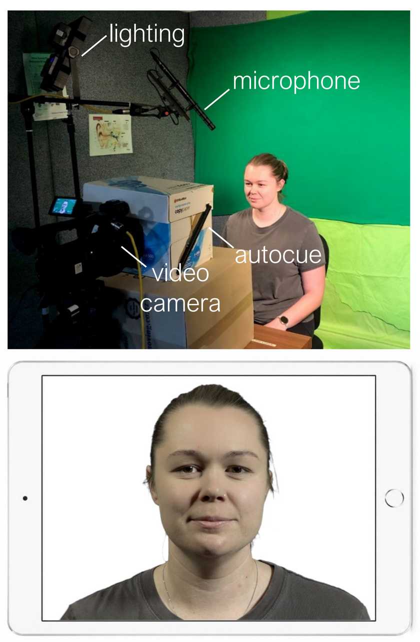


Figure 1: Audiovisual recording process. Setup (top) and post-processed frame.

Results:

- Strong image agreement was observed for several target words, including: chalk (98%), comb (95%), cap (86%), dirt (81%), leg (81%).
- Low-pass-filter results informed later item selection

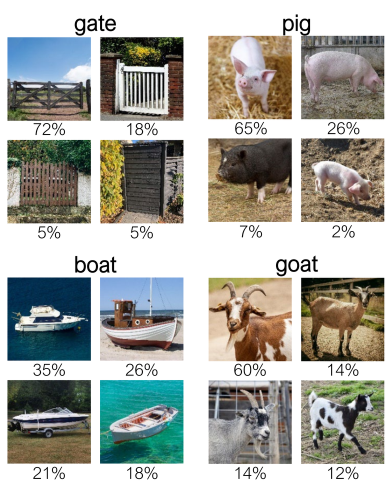


Figure 2 *(left)*: Example vote distributions for candidate images corresponding to the target words *gate*, *pig*, *boat*, and *goat*.

Figure 3 *(right)*: A finalised UC4AFC test plate for the target word *goat* with foils *boat*, *gate*, and *pig*.

Study 2 – Spontaneous naming by 4-year-olds (Chong, 2025)

Objectives:

i) Evaluate whether four-year-olds consistently produced intended labels

ii) Quantify naming variability and naming latency

Method:

- 30 four-year-old children from low socioeconomic backgrounds participated
- Children viewed:
  - 98 selected UC4AFC images, 17 alternative candidate images
- Images presented with prompting phrases ("This is…", "These are…", etc.)
- Naming latency measured using synchronised audiovisual recordings
- Analyses included:
  - intended name agreement (the response matched our intended name)
  - target word inclusion (the target word was included within a longer response)
  - naming diversity using entropy statistic H (Snodgrass & Vanderwart, 1980) expressed as a percentage of the theoretical maximum for the sample size

Results:

Naming diversity ranged from near 0% (high agreement/low variability) to ~90% of theoretical maximum entropy (low agreement/high variability).

Figure 4: Intended name agreement and naming diversity

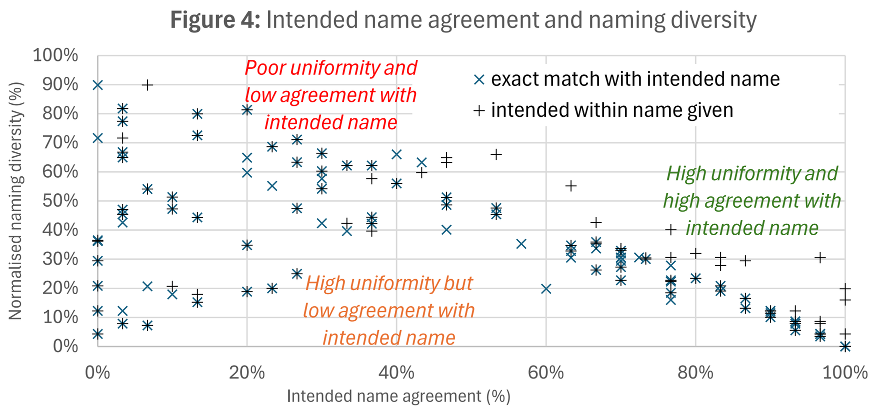
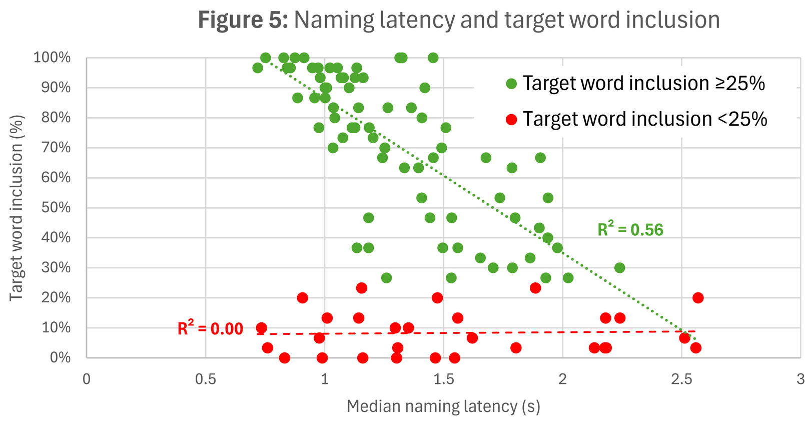


Figure 5: Naming latency and target word inclusion



Study 3 – Receptive identification by 4-year-olds (Riley, 2026)

Objective: Determine whether four-year-olds could accurately and rapidly identify intended targets in 4AFC format

Method:

- 45 four-year-old children completed receptive 4AFC task
- Accuracy and response latency recorded for all 98 target items

Results:

- Approximately three quarters of items demonstrated:
  - >75% identification accuracy
  - median latency <3.5 s
- These items identified as strong candidates for final inclusion
- Poorer-performing items considered for exclusion

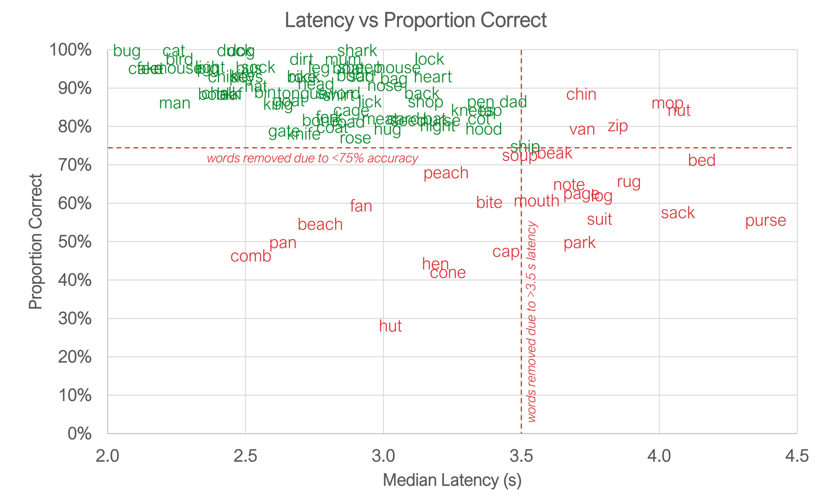


Figure 6: Median receptive identification latency versus proportion correctly identified for each target item.

Final list composition

Final inclusion decisions incorporated:

- low-pass filter normalisation results
- expressive naming results
- receptive identification results
- phonemic foil balance requirements

Item #	Target	Foil 1	Foil 2	Foil 3	Target and 1	Target and 2	Target and 3
1	**lock**	sock	lick	knees	V, FINAL C	INITIAL C, FINAL C	UNRELATED
2	**kite**	night	cot	chalk	V, FINAL C	INITIAL C, FINAL C	SEMANTIC
3	**coat**	boat	kite	sock	V, FINAL C	INITIAL C, FINAL C	SEMANTIC
4	**bat**	hat	back	duck	V, FINAL C	INITIAL C, V	SEMANTIC
5	**keys**	knees	sheep	house	V, FINAL C	V	SEMANTIC
6	**feet**	meat	seed	nose	V, FINAL C	V	SEMANTIC
7	**shirt**	dirt	shark	bin	V, FINAL C	INITIAL C	UNRELATED
8	**dirt**	shirt	duck	card	V, FINAL C	INITIAL C	UNRELATED
9	**fork**	chalk	feet	nose	V, FINAL C	INITIAL C	UNRELATED
10	**sad**	dad	sock	light	V, FINAL C	INITIAL C	UNRELATED
11	**bin**	chin	bug	shark	V, FINAL C	INITIAL C	UNRELATED
12	**house**	mouse	head	gate	V, FINAL C	INITIAL C	SEMANTIC
13	**sock**	rock	seed	boot	V, FINAL C	INITIAL C	SEMANTIC
14	**boat**	coat	bike	shark	V, FINAL C	INITIAL C	SEMANTIC
15	**chip**	chin	chalk	nurse	INITIAL C, V	INITIAL C	UNRELATED
16	**night**	knife	coat	leaf	INITIAL C, V	FINAL C	UNRELATED
17	**back**	bag	bike	shirt	INITIAL C, V	INITIAL C, FINAL C	SEMANTIC
18	**heart**	card	hat	nose	V	INITIAL C, FINAL C	SEMANTIC
19	**seed**	keys	sword	dirt	V	INITIAL C, FINAL C	SEMANTIC
20	**book**	hood	bike	heart	V	INITIAL C, FINAL C	UNRELATED
21	**duck**	bus	dog	feet	V	INITIAL C, FINAL C	UNRELATED
22	**gate**	cake	goat	bone	V	INITIAL C, FINAL C	UNRELATED
23	**sword**	fork	sad	mum	V	INITIAL C, FINAL C	UNRELATED
24	**card**	heart	bird	leaf	V	FINAL C	UNRELATED
25	**road**	boat	sword	shirt	V	FINAL C	UNRELATED
26	**pig**	bin	leg	sheep	V	FINAL C	SEMANTIC
27	**tongue**	bug	king	chip	V	FINAL C	SEMANTIC
28	**dog**	rock	dad	sheep	V	INITIAL C	SEMANTIC
29	**pen**	leg	pig	chalk	V	INITIAL C	SEMANTIC
30	**leaf**	knees	lock	man	V	INITIAL C	UNRELATED
31	**man**	sad	mouse	shark	V	INITIAL C	UNRELATED
32	**mum**	bus	mouse	lock	V	INITIAL C	UNRELATED
33	**leg**	pen	lick	shop	V	INITIAL C	UNRELATED

Item #	Target	Foil 1	Foil 2	Foil 3	Target and 1	Target and 2	Target and 3
1	**bug**	hug	bag	hood	V, FINAL C	INITIAL C, FINAL C	UNRELATED
2	**goat**	boat	gate	pig	V, FINAL C	INITIAL C, FINAL C	SEMANTIC
3	**cat**	hat	cot	bird	V, FINAL C	INITIAL C, FINAL C	SEMANTIC
4	**chin**	bin	chip	back	V, FINAL C	INITIAL C, V	SEMANTIC
5	**knees**	keys	leaf	head	V, FINAL C	V	SEMANTIC
6	**ship**	chip	lick	hug	V, FINAL C	V	UNRELATED
7	**hat**	bat	house	lick	V, FINAL C	INITIAL C	UNRELATED
8	**hug**	bug	hat	seed	V, FINAL C	INITIAL C	UNRELATED
9	**rock**	sock	road	boot	V, FINAL C	INITIAL C	UNRELATED
10	**meat**	feet	mouse	dad	V, FINAL C	INITIAL C	UNRELATED
11	**light**	kite	leg	house	V, FINAL C	INITIAL C	SEMANTIC
12	**dad**	sad	dirt	mum	V, FINAL C	INITIAL C	SEMANTIC
13	**chalk**	fork	chin	pen	V, FINAL C	INITIAL C	SEMANTIC
14	**mouse**	house	man	cat	V, FINAL C	INITIAL C	SEMANTIC
15	**bone**	boat	book	tap	INITIAL C, V	INITIAL C	UNRELATED
16	**bag**	bat	hug	meat	INITIAL C, V	FINAL C	UNRELATED
17	**boot**	bag	bat	feet	INITIAL C	INITIAL C, FINAL C	SEMANTIC
18	**sheep**	meat	shop	dog	V	INITIAL C, FINAL C	SEMANTIC
19	**bike**	light	book	road	V	INITIAL C, FINAL C	SEMANTIC
20	**lick**	ship	lock	sword	V	INITIAL C, FINAL C	UNRELATED
21	**cot**	rock	cat	bone	V	INITIAL C, FINAL C	UNRELATED
22	**shop**	cot	ship	head	V	INITIAL C, FINAL C	UNRELATED
23	**bus**	duck	nurse	tap	V	FINAL C	UNRELATED
24	**hood**	book	bird	knife	V	FINAL C	UNRELATED
25	**nurse**	bird	bus	seed	V	FINAL C	UNRELATED
26	**nose**	coat	keys	tongue	V	FINAL C	SEMANTIC
27	**king**	bin	tongue	sword	V	FINAL C	SEMANTIC
28	**knife**	light	night	cake	V	INITIAL C, V	SEMANTIC
29	**shark**	heart	shop	goat	V	INITIAL C	SEMANTIC
30	**cake**	gate	king	tongue	V	INITIAL C	SEMANTIC
31	**tap**	back	tongue	dog	V	INITIAL C	UNRELATED
32	**head**	leg	heart	boot	V	INITIAL C	UNRELATED
33	**bird**	nurse	boot	tap	V	INITIAL C	UNRELATED

Table 1: The two finalised UC4AFC word lists, and the categories of each foil.

Category	Set 1	Set 2	Total
Initial C only	15	15	30
Initial C, V	4	4	8
V only	18	18	36
Final C only	5	6	11
V, final C	14	14	28
Initial C, final C	10	9	19
Semantic	15	16	31
Unrelated	18	17	35
Initial C total	29	28	57
V total	36	36	72
Final C total	29	29	58

Vowel	Set 1	Set 2	Total Targets	Targets %
TRAP	4	5	9	13.6%
FLEECE	4	3	7	10.6%
KIT	3	4	7	10.6%
LOT	3	3	6	9.1%
STRUT	3	3	6	9.1%
GOAT	3	3	6	9.1%
PRICE	2	3	5	7.6%
NURSE	2	2	4	6.1%
THOUGHT	2	1	3	4.5%
START	2	1	3	4.5%
DRESS	2	1	3	4.5%
MOUTH	1	1	2	3.0%
FOOT	1	1	2	3.0%
FACE	1	1	2	3.0%
GOOSE	0	1	1	1.5%
Total	33	33	66	100%

Tables 2 and 3: Occurrences of foil categories (above) and target vowels (right) in the two finalised word lists.

- No included target words fell below 75% correct on receptive testing
- Only *chin* and *ship* exceeded 3.5 s median receptive latency
- Poor spontaneous naming did not necessarily predict poor receptive identification
  - Ten included items had <25% expressive target-word inclusion but still achieved 85% ± 5% receptive accuracy
  - Expressive and receptive latencies showed weak but significant correlation ($r = .35$, $p < .01$)

Conclusion

- The UC4AFC provides robust four-alternative monosyllabic stimuli that are more suitable for New Zealand and Australian children than previous North American materials.
- These findings support its use in paediatric speech discrimination assessment and future APD evaluation.

References

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COI: None declared